

Myofascial continuity: Towards a new understanding of human anatomy

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Introduction

Since time immemorial, people have strived to understand the fascinating anatomy of the human body. From Herophilus of Chalcedon (325–255 BC), who performed the first systematic dissections in Alexandria, until today, categorization and isolation represent the most important intellectual tools used to explain our inherent complexity: for instance, we distinguish a series of organ systems, all suggested to exert a specific set of functions. With regard to the locomotor system, diligent cadaver dissections have identified a total of 556 skeletal muscles. Medical students learn their names as well as their discrete origins and insertions, observing them in pictures in anatomy textbooks. However, when instructed to expose a specific tissue and its boundaries in dissection courses, they often discover that the human body, in reality, looks different to the brightly colored illustrations. Indeed, pinpointing the beginning and the end of a structure is hardly possible for the unaided eye: the scalpel needs to artificially create these points.

The isolated consideration of the skeletal muscles has severe consequences for the assumed biomechanics. As an example, the latissimus dorsi muscle, following the classical conventions, originates in the lumbar spine and inserts in the crista tuberculi minoris humeri, a prominence of the upper arm bone. If this were true, the cranial forces developed upon contraction or stretching of the latissimus would exclusively be transmitted to

the humerus. However, cadaveric studies clearly demonstrate that the distal attachment of the muscle is far more complex. In addition to the bony insertions, some muscular fibers fuse strongly with the posterior aspect of the brachial fascia, which wraps around the muscles of the upper arm (Stecco et al., 2007; 2008), and in turn connects to the antebrachial fascia of the lower arm (Stecco et al., 2007). This example illustrates that contractile forces developed in the latissimus—based on its true anatomical course—may even affect the soft tissue of the hand (for an overview of the evidence regarding myofascial force transmission, see Chapters 13 and 14).

Concepts of myofascial chains

Decades ago, a number of practitioners raised doubts about the structural independence of neighboring body regions, and suggested the existence of so-called myofascial chains. A common feature of the different systems is the proposed morphological linkage between the skeletal muscles: in opposition to the idea of muscle chains or muscle slings (Tittel, 2003), where muscles are grouped based on a common function within a certain movement (e.g. extensor chain of the leg: gastrocnemius, quadriceps femoris and gluteus maximus), the components of myofascial chains are structurally connected. As the term “myofascial” indicates, a variety of tissues (“myo” for muscle, “fascial” for the diverse components of the fascial system) can form the involved continuities.

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Françoise Mézières (1909–1991) can be considered one of the earliest and most important pioneers who presented a system of myofascial chains (*chaînes musculaires*). The French therapist described four major lines of soft tissue continuity. Her most important, the *chaîne postérieure*, was first reported in 1949 and included a deep and a superficial plane extending dorsally from the head to the toes. In addition, Mézières proposed a deep anterior chain consisting of the continuity between the iliopsoas and the diaphragm, a brachial chain (biceps brachii, brachialis, brachioradialis, pronator teres/quadratus, flexor carpi ulnaris/radialis) and the anterior neck chain (longus colli, longus capitis and rectus capitis anterior). Although she did not provide an all-encompassing system for all parts of the body (e.g. there is no lateral chain) and her lines had either an astonishingly high (posterior) or small (deep anterior, anterior neck) number of components, Mézières' works paved the way to what can be called the age of myofascial chains. In the 1990s, several other authors presented their ideas about how the body might be connected (e.g. Busquet, 1992; Denys-Struyf, 1995; Myers, 1999). The different suggested concepts overlap significantly in a series of aspects. The connection between the psoas muscle and the diaphragm, first proposed by Mézières, is also included in the superficial front line of Myers. Likewise, both Busquet and Myers designed similarly constructed diagonal chains on the dorsal and ventral side of the body, connecting the arms to the opposite legs. However, the authors also introduced their own innovations. Besides five myofascial chains, Léopold Busquet additionally described a visceral and a neurovascular chain. Godelieve Denys-Struyf, a Belgian physiotherapist and osteopath, presented a mix of myofascial chains alongside postural patterns. Finally, Thomas Myers, who suggested a system

of 11 muscle-fascia lines, was particularly interested in linear connections between the components of chains. For instance, his superficial back line includes a structural linkage from the back extensor muscle/the lumbar fascia to the hamstrings but not to the gluteus maximus muscle because with the latter included, the direction of continuity would deviate more strongly from the cranio-caudal course of the chain. This is of relevance from a mechanical perspective because on a linear course, compared with a curved one, more force can be transferred.

Evidence for the existence of myofascial chains

The recent discoveries from fascia research have popularized treatments according to myofascial chains. Practitioners working according to the proposed concepts aim to evoke effects remote to the area of symptom localization, assuming that local dysfunctions and problems such as flexibility impairments, increased stiffness or muscle hypertonicity may (1) radiate towards or (2) originate from distant structures via the body-wide network of tissue continuity. These ideas are intriguing and revolutionary as they offer the possibility to explain and alleviate many pain conditions where local treatments have failed. Nonetheless, scientific proof for the existence of myofascial chains was, for a long time, not forthcoming, and thus all treatments performed were solely based on the clinical experience, self-reported dissection work, and beliefs of the opinion leaders who were developing the different concepts. During the last decade, however, the research output regarding this topic has increased substantially.

A comprehensive systematic review (Wilke et al., 2016) examined the question as to whether basic anatomical cadaver studies support the existence

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of six of the eleven myofascial chains described by Myers (see Figure 12.1). The authors found convincing evidence for the posterior longitudinal chain, which consists of the plantar aponeurosis, Achilles tendon, gastrocnemius, hamstrings, sacrotuberous ligament, lumbar fascia and back extensor muscle. In addition, two diagonal connections between the upper and lower limb were verified. On the frontal side of the body (anterior diagonal chain), fascial tissue connects the arm via the pectoralis major and the rectus abdominis with the opposite leg (adductor longus). A similar line (posterior diagonal chain) extends

on the back side, including the latissimus dorsi, the lumbar fascia and the contralateral gluteus maximus muscle. Despite these findings, not all of the lines were confirmed: while only parts of the lateral and spiral-like chain were shown to exist, there was no evidence for an anterior longitudinal chain.

Although recent research has corroborated the existence of at least some (parts) of the muscle-fascia lines (Table 12.1), there are various important key points to consider. First, imagining myofascial chains as the only pathways

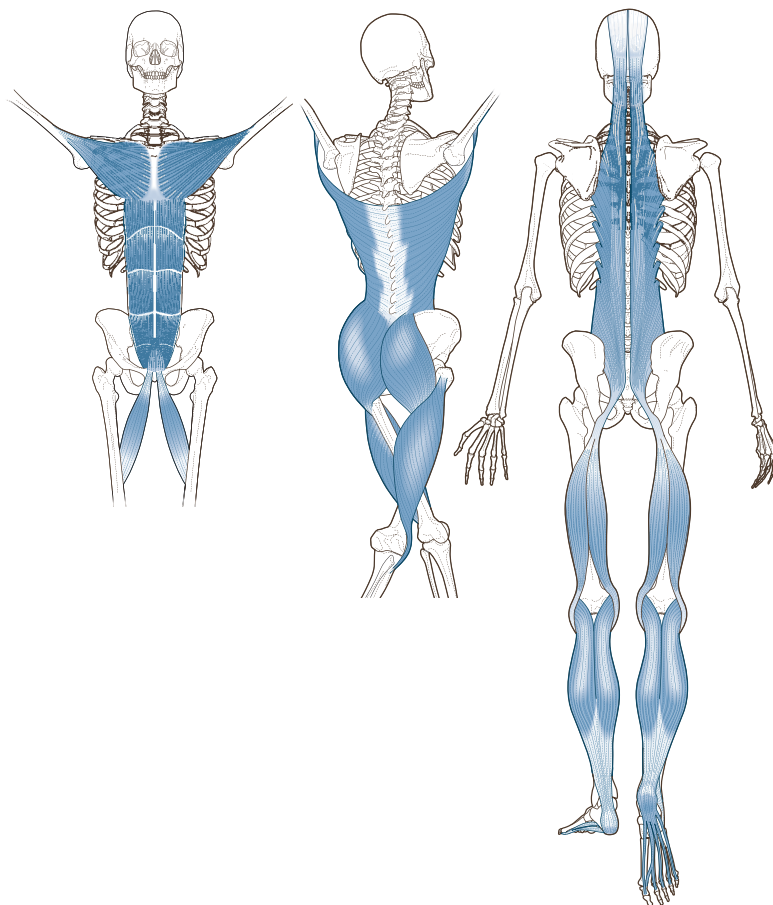


Figure 12.1

Evidence is solid for the existence of three myofascial chains: the anterior diagonal chain (left), posterior diagonal chain (center) and the posterior longitudinal chain (right). Note: the diagonal chains are bilateral, extending (1) from the left arm to the right leg and (2) vice versa.

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Table 12.1 Examples of myofascial chains and their assumed function (only verified continuities are listed)

Myofascial chain	Soft tissue components	Clinical comments
Lateral longitudinal chain	• Crural fascia • Iliotibial tract • Gluteus maximus muscle/tensor fasciae latae muscle • External/internal obliquus abdominis muscle	Helps to stabilize the body (ankle, knee, hip and spine) in the frontal plane. Hypothesized continuation between ribcage and head is not yet confirmed
Spiral chain	• External/internal obliquus abdominis muscle • Serratus anterior muscle • Rhomboideus muscles	Engaged in rotatory movements (e.g. throwing). Hypothesized continuations in cranial and caudal direction not yet confirmed
Posterior longitudinal chain	• Plantar aponeurosis • Achilles tendon • Gastrocnemius muscle • Hamstring muscles • Sacrotuberous ligament • Lumbar fascia and erector spinae muscle	Provides resistance against gravity and helps to maintain upright posture. Of potential relevance in linear vertical movements (e.g. jumps)
Posterior diagonal chain	• Vastus lateralis muscle • Gluteus maximus muscle • Lumbar fascia • Contralateral latissimus dorsi muscle • Brachial fascia	Dorsal connection of both body sides. May be important in unilateral movements requiring arm-leg coupling
Anterior diagonal chain	• Adductor longus muscle • Rectus abdominis muscle • Contralateral pectoralis major muscle • Brachial fascia	Ventral connection of both body sides. May be important in unilateral movements requiring arm-leg coupling (e.g. throwing, serving when playing tennis, kicking a football)

potentially transmitting relevant mechanical forces is a fallacy. Tissue continuity is not restricted to the connected structures grouped together; it also affects other parts not included in a certain chain. For instance, the superficial back arm line from Myers describes a connection between the deltoideus muscle and the trapezius muscle. Although this morphological linkage has in fact been verified by cadaveric research, there is also a continuity from the deltoid fascia to that of the rhomboideus muscle (Day et al., 2008). It is hence evident that myofascial chains

can take different tracks on their way through the body. Second, besides displaying connections in an in-series arrangement, extensive research (although mainly carried out in animals) has demonstrated most muscles to additionally fuse with others located parallel to them. As an example, in the lower leg, the fascia of the tibialis anterior merges with the connective tissue surrounding synergists such as the extensor hallucis longus and even mechanically interacts with distant antagonists such as the triceps surae complex (Rijkkelijkhuisen et al., 2007; Yucesoy, 2010).

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Myofascial continuity therefore represents a global construction principle of the body rather than a specific hallmark of muscle-fascia chains. Finally, the strength of a myofascial continuity is not constant and can vary depending on several factors: this is best established for the connection between the plantar aponeurosis (PA) and the Achilles tendon (AT). In a cadaver study, Snow et al. (1995) investigated the feet of adults, neonates and fetuses. Analyses of the heel region revealed a strong fiber continuity between the aponeurosis and the tendon when considering only the youngest specimens. However, in people in their 20s, only superficial fibers of the periosteum formed the tissue bridge between both structures. No direct continuity was observed in the elderly as the PA and the AT presented only a bony insertion at the calcaneus. The age-dependency found by Snow et al. was confirmed in subsequent studies. Ballal et al. (2014) dissected 12 adult feet: only in the two youngest (aged 59 and 66 years) and a 79-year-old cadaver, were the AT and the PA clearly fused. Kim et al. (2010) investigated a total of 60 lower extremities of cadavers with a mean age of 68 (43–98) years. Morphological fusion of the AT and the PA (found in 8% of the cases) correlated with the age of the specimens. In a follow-up study using magnet resonance imaging (MRI), the same authors examined the insertion of the tendon as a function of age in younger specimens. Interestingly, they found a progressive migration into the proximal direction: the older the individuals, the more superior the insertion (Kim et al., 2011). The most recent study describing the connection between the aponeurosis and the tendon was published by Stecco et al. (2013). Contrary to previous reports, the authors, who were the only ones to use fresh cadavers without fixation, detected a direct morphological linkage in all of the 12 examined cadavers: at the heel, the PA was

reported to continue as a thin band (thickness: 1–2 mm), blending with the paratenon of the AT.

In sum, the results regarding the myofascial continuity at the calcaneus are conflicting. The majority of studies suggest an age-related decrease or disappearance of the tissue linkage. However, it is also possible that in this subsample, the MRI technique (Kim et al., 2010; 2011) was not sensitive enough to detect small degrees of continuity and that the use of embalmed cadavers (Ballal et al., 2014; Snow et al., 1995) may partly have made it impossible to identify these due to adhesions between periosteal fibers and bones.

Summary

The frequently conveyed concept of skeletal muscles as being independent actuators does not sufficiently reflect the complexity of human anatomy. Both the skeletal muscles, as well as the surrounding and neighboring connective tissues, are directly fused with each other, creating a body-wide web of structural continuity. Current evidence has verified several connections that can be grouped to myofascial chains. Such muscle-fascia lines may explain the radiation of pain and dysfunction to remote body locations. However, reaching in multiple directions, structural linkage is by no way restricted to these tracks. Reducing the construct of tissue continuity to the proposed chains would therefore create the same artificial boundaries as those portrayed by classical anatomy books defining the discrete origins and insertions of the skeletal muscles. Consequently, when evaluating the mechanical interactions between adjacent tissues (Chapters 13 and 14), it is imperative to focus on both the potential clinical relevance of force transmission in series and to structures located in parallel.

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